

Ariana Ariaei

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RESEARCH SUMMARY

My research focuses on applying deep learning and multimodal AI models to healthcare diagnostics and behavioral analysis. I am particularly interested in developing machine learning solutions for medical imaging, cardiovascular disease detection, and stress response analysis using physiological signals and computer vision techniques.

EDUCATION

2022 - present Bachelor of Science in Computer Engineering at **Babol Noshirvani University of Technology** (Ranked 501 – 600th in [World University Rankings 2025](#))
GPA: 3.6/4.0 over last 2 years
Expected Graduation: July 2026

RESEARCH EXPERIENCE

Research Assistant, Artificial Intelligence Research Lab *2025–present*
Babol Noshirvani University of Technology
Working on machine learning applications in signal processing and software systems.

RESEARCH INTERESTS

Machine Learning, Deep Learning, Computer Vision, Signal Processing, Pattern Recognition, Large Language Models (LLMs), and AI applications in Healthcare and Medical Imaging.

PUBLICATIONS

The Role of Machine Learning in Diagnosis of Acute Occlusion Myocardial Infarction in Presence of Left Bundle Branch Block (LBBB) *In Progress*
Under supervision of Dr. H. Omranpour. Developing machine learning models to automate detection of myocardial infarction in presence of LBBB using ECG data. Implementing deep learning algorithms for classification and exploring novel diagnostic criteria. Manuscript in preparation for submission to medical imaging journal.

AI-Driven Multimodal Analysis of Stress Responses in Virtual Reality Simulations *In Progress*
Under supervision of Dr. H. Omranpour. Developed VR environments in Unity and collected multimodal data (EEG, ECG, EMG, facial expressions, postures). Applying deep learning and LLMs for behavioral pattern recognition to uncover correlations between physiological responses and decision-making under stress.

ACADEMIC PROJECTS

Scam Sleuth: AI-Powered Scam Reporting and Detection Platform *2024–2025*
Designed an AI-powered scam detection platform using Python, .NET, and Next.js. Implemented secure

authentication, trust-score algorithms for website analysis, admin dashboard for report management, and user engagement features including comments and blog posts.

Tslator: A Compiler for TSLANG

2025

Built compiler for educational C-like language (TSLANG) using Python and PLY. Implemented lexical analysis, parsing, semantic analyzer with scoped symbol tables, type checking, AST construction, and intermediate representation (IR) code generation. Supports functions, arrays, conditionals, loops, and string handling.

PyThreadServe: Multi-Process HTTP/1.0 Server with Concurrency Control

2025

Implemented lightweight HTTP/1.0 server in Python with multi-process architecture and thread pooling. Features include GET/POST support, round-robin load balancing, thread-safe file operations, directory traversal protection, and comprehensive logging with configurable concurrency limits.

TEACHING EXPERIENCE

- **Teaching Assistant for Principles of Compiler Design**

2025

Instructed by: Dr. A. Gholami Rudi

Assisted with grading assignments, conducting problem-solving sessions, and providing support for compiler implementation projects.

- **Teaching Assistant for Applied Linear Algebra**

2024

Instructed by: Dr. J. Kazemitabar

Held weekly recitation sessions, graded homework and exams, and assisted students with matrix operations and applications.

- **Teaching Assistant for Data Structures**

2024

Instructed by: Dr. H. Omranpour

Supervised laboratory sessions, evaluated coding assignments, and provided guidance on algorithm implementation.

- **Teaching Assistant for Introduction to Computer Science and Programming**

2023

Instructed by: Dr. H. Omranpour

Conducted tutorial sessions for introductory programming concepts, graded assignments, and mentored students in C++.

AWARDS AND HONORS

- **Ranked among top 6%** ($\approx 10,000$ of $150,000+$) in Iranian University Entrance Exam, one of the most competitive academic assessments nationally.
- **Full tuition scholarship** for B.Sc. in Computer Engineering at Babol Noshirvani University of Technology (merit-based, covering all tuition fees).
- **Ranked top 6%** (10 of 167 participants) in the 6th AlgoNIT National Programming Contest, demonstrating strong problem-solving and competitive programming skills.
- **Ranked 1st at high school** in the Iranian University Entrance Exam among fellow students specializing in Mathematics and Physics.

SELECTED COURSES

Principles of Compiler Design

Score: 19.6/20

Covered lexical analysis, parsing, semantic analysis, and intermediate code generation.

Fundamentals and Applications of Artificial Intelligence

Score: 17.1/20

Explored search algorithms, knowledge representation, machine learning fundamentals, and AI applications.

Formal Languages & Automata Theory

Score: 18.5/20

Studied regular languages, context-free grammars, pushdown automata, and Turing machines.

Computer Architecture

Score: 18/20

Covered processor design, memory hierarchies, pipelining, and performance optimization.

Applied Linear Algebra

Score: 19.25/20

Focused on matrix operations, eigenvalues, vector spaces, and applications to data science.

Software Systems Analysis and Design

Score: 19.6/20

Covered requirements engineering, UML modeling, design patterns, and software architecture.

SKILLS

Computer Skills

- **Programming Languages & Frameworks:** Python, C#, .NET, C++, VHDL, Matlab, Shell Script
- **Machine Learning Libraries:** PyTorch, TensorFlow, Pytesseract, NumPy, Pandas, Scikit-learn, OpenCV, Matplotlib, LangChain
- **Technologies:** Unity Engine, Docker, Git, LaTeX

Language Skills

- **English:** Preparing for IELTS exam (scheduled for November 14th, 2025)

EXTRACURRICULAR ACTIVITIES

Volunteer Staff Member, 5th AlgoNIT National Programming Contest

2024

Supervised examiners, coordinated contest logistics, and assisted with technical support during the national programming competition.

REFERENCES

Dr. Ali Gholami Rudi

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Dr. Seyed Javad Kazemitabar

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Dr. Hesam Omranpour

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